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Searching for Planets around Altair with JWST/NIRcam

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ABSTRACT

Altair (α Aql) is one of the nearest stars to the Sun at 5.13 pc and its age has recently been revised to 88±10 Myr, making it one of the youngest stars in the Solar neighborhood. This makes Altair a compelling target for high-contrast direct imaging searches for giant planets. We present two epochs of simultaneous F444W and F200W MASK430R JWST/NIRCam coronagraphic imaging observations. After post-processing, these data achieve F444W contrast levels of 2 x 10^{-6} at 1" (5.1 au), 1.5 x 10^{-7} at 2" (10.3 au), and a few x 10^{-8} beyond 4"(20 au). Using the Sonora Flame Skimmer models, this corresponds to sub-Jupiter mass sensitivity beyond 5 au and sub-Saturn mass sensitivity beyond 10 au. We identified several candidate point sources in the field in the first epoch data. Second epoch observations confirmed that none of the candidates were co-moving with Altair and their colors are consistent with background stars. Altair has no known companions, no detected debris disk, and is not associated with any stellar association. Despite this, its youth and proximity ensure it will remain a high-priority target for future high-contrast imaging observations. We additionally discuss Altair as a potential reference star for future observations, presenting measurements of background sources in the field and showing that Altair will remain at least a few arcsec from background stars over the next 10–20 years, making it a reliable reference target for upcoming imaging campaigns including the Roman Space Telescope Coronagraph Instrument.

Keywords: Exoplanets (498) — Exoplanet systems (484) — Coronagraphic imaging (313) – James Webb Space Telescope (2291) – Extrasolar gaseous giant planets (509)

1. INTRODUCTION

The bright (V = 0.77) A7V-type star Altair (α Aql, HD 187642, HIP 97649) is the second nearest (ϖ = 194.95 ± 0.57 mas, d = 5.130 ± 0.015 pc) A-type star after Sirius (Mermilliod 1997; Gray et al. 2003; van Leeuwen 2007). Thus far, Altair has no known stellar or substellar companions. Its rapid rotation (v sin i =221 km s^{-1}, Erspamer & North 2003) and extreme brightness makes radial velocity and transit planet searches impractical, and no companion has been detected in interferometric observations (Monnier et al. 2007; Bouchaud et al. 2020). Coronagraphic imaging remains a promising method to discover companions at wider separations. Despite having an entry in the Washington Double Star Catalog (WDS J19508+0852A) and having a half dozen associated companions identified B through G with separations ranging from 27″ to 196″, all appear to be faint, low proper motion background

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Andrea's notes

1 1pc = the distance to an object that has a parallax angle of 1 arcsecond.
1 arcsecond = 1/3600 of a degree

2 These are the names for different filters on JWST!

3 A coronagraph is an instrument that blocks out the starlight to see the MUCH fainter light reflected off a planet, or the glow of a really young hot one.

4 this means that any planet with a Jupiter mass or more would be detectable beyond 5 AU and likewise for Saturn beyond 10 AU.

5 This highlights the importance of thorough verification/confirmation of results! It also shows that it is valuable to report a "null" result, as this might help other scientists interested in Altair to maybe redirect their efforts elsewhere!

6 stars have what we call proper motion - which means they also slowly move across the night sky over time. Given that it is a lone star without contaminant background stars for a long time it will be a good reference for calibration measurements for example.

stars inconsistent with Altair's proximity or motion (see subsection 5.1).

Altair is a δ Scuti pulsator and one of the most rapidly rotating stars in the solar neighborhood (e.g. van Belle et al. 2001; Buzasi et al. 2005). It rotates at ∼74% of its angular (breakup) velocity with an equatorial velocity likely exceeding 300 $\mathrm{km\,s^{-1}}$ (Rieutord et al. 2023). This extreme spin rate flattens the star: interferometric observations directly resolved this oblateness, finding the equatorial radius ∼22% larger than the polar radius, and produced the first resolved image of Altair, revealing strong gravity darkening (Monnier et al. 2007). A 2D structural model using the ESTER code (Bouchaud et al. 2020) simultaneously fit these interferometric constraints alongside spectroscopic and asteroseismic data, yielding a mass of $1.86 \pm 0.03\ M_\odot$ and showed that the core rotates ∼50% faster than the envelope.

Most nearby, A-type stars (i.e., Vega and Fomalhaut) have been extensively studied for numerous imaging searches for planets from both ground and space based telescopes (Heinze et al. 2008; Janson et al. 2015; Meshkat et al. 2018; Ren et al. 2023; Kalas et al. 2008; Ygouf et al. 2024; Beichman et al. 2025a). Altair, however, has largely been overlooked from deep targeted analyses like Vega and Fomalhaut due to its assumed older age from 1D stellar evolution models (1.2–1.4 Gyr; Domiciano de Souza et al. 2005). A recent analysis of TESS data using 2D structural modeling (Rieutord et al. 2024) derived a younger age for Altair than previous estimates: 88 ± 10 Myr. Altair is now the closest A-star with an age under 100 Myr, making it an ideal and exciting new candidate for direct imaging searches for young giant exoplanets. JWST coronagraphy is particularly well suited for this search, as its deep infrared sensitivity enables access to sub-Jupiter mass companions around young stars (Carter et al. 2021, 2023), as recently demonstrated by the discovery of a sub-Jupiter mass planet TWA 7b (Lagrange et al. 2025; Crotts et al. 2025)

Despite its youth, Altair shows no evidence for a debris disk. Photometric observations at 24 μm show no significant infrared excess (1%; Rieke et al. 2005), and resolved imaging in the near-infrared and submillimeter also show no disk detection (Thureau et al. 2014; Spalding et al. 2022). The lack of a detected disk around Altair is surprising since its revised age (88 ± 10 Myr) puts it comfortably in the range when debris disks around A-type stars are most commonly detected (tens to hundreds of Myr; Wyatt 2008). Notably, near-infrared interferometry with CHARA/FLUOR does reveal a $3.07 \pm 0.24\%$ K-band excess around Altair, attributed to an extended circumstellar source (Absil et al. 2013), suggesting some circumstellar material may persist in the inner system.

Altair's brightness, lack of a debris disk, and absence of known companions also make it a promising candidate as a reference star for future high-contrast coronagraphic imaging campaigns, both from the ground and from space potentially with Roman Coronagraph Instrument (Bailey et al. 2023; Hom et al. 2026), further motivating a thorough characterization of its circumstellar environment in this work with JWST/NIRCam.

2. OBSERVATIONS

We observed Altair with JWST/NIRCam on 2024 October 11 as part of the NIRCam GTO program (PI. Beichman, Program ID #4534)[10]. Altair was observed simultaneously in the long (F444W, 4.44 μm) and short (F200W, 2.0 μm) wavelength channels using the M430R round Lyot coronagraph, which has an inner working angle of $0\farcs43$ in subarray mode ($\pm10''$ field of view). This choice of mask minimizes the effects of saturation at the modest expense of inner angle ($<1''$). We observed Altair at two roll angles separated by ∼8 ° to enable Angular Differential Imaging (ADI; Marois et al. (2006); Liu (2004)). Each roll consisted of 650 integrations of 5 groups in RAPID readout mode, for a total integration time of 4182 s per roll and a total on sky integration time of 8364 s (∼2.3 hr) after combining both rolls. We also observed Vega as a reference star for perform Reference Differential Imaging (RDI) using the same filter and mask configuration, with 5 dithered pointings of 140 integrations each (700 total integrations). Both targets were acquired using the F335M filter with the neutral density square. Table 2 summarizes the observing parameters.

By observing Altair in both F444W and F200W, we were able to reject out any detected sources as background stars, since stars are significantly brighter in 2 μm than in 4.44 μm. However, since the M430R mask only covers one half of the sky in F200W, we only can rule out detected companions to the south of F444W (see Figure 1).

We obtained second epoch observations of Altair with NIRCam on 2025 May 26 (PI Meshkat, Program ID #8507). We used the same filters, roll angles, reference star, masks and integration times, in order to have as comparable observations as possible, with the primary difference that the observation was scheduled at a time where the M430R mask covered the sky to the North of Altair for the F200W images.

[10] All the JWST data used in this paper can be found in MAST: 10.17909/61yw-dq54.

Table 1. Properties of the Altair

Property	Value	Units	Comments
Spectral Type	A7V		Gray et al. (2003)
Apparent V_{mag}	0.765 ± 0.015	mag	Mermilliod (1997)
Apparent M_{mag}	0.245 ± 0.025	mag	Bouchet et al. (1991)
T_{pole}	8450±140	K	Monnier et al. (2007)
T_{eq}	6860±150	K	Monnier et al. (2007)
Mass	$1.86^{+0.03}_{-0.03}$	$M_\odot$	Bouchaud et al. (2020)
$\log(L/L_\odot)$	1.027±0.011	dex	Peterson et al. (2006)
Age[a]	88±10	Myr	Rieutord et al. (2024)
[Fe/H]	-0.2	dex	Monnier et al. (2007)
$\log(g)$	4.266±0.012	cgs	Peterson et al. (2006)
inclination (i)	$50°.65 \pm 1°.23$		Bouchaud et al. (2020)
θ_{pole}	2.958±0.020	mas	Monnier et al. (2007)
θ_{eq}	3.670±0.014	mas	Monnier et al. (2007)
α_{ICRS} (epoch J2000)	$19^h50^m47.00^s$		van Leeuwen (2007)
δ_{ICRS} (epoch J2000)	$+08°52'05.96''$		van Leeuwen (2007)
α_{ICRS} (epoch J2024.78)	$19^h50^m47.88^s$		This work
δ_{ICRS} (epoch J2024.78)	$+8°52'15.09''$		This work
α_{ICRS} (epoch J2025.40)	$19^h50^m47.92^s$		This work
δ_{ICRS} (epoch J2025.40)	$+8°52'16.18''$		This work
Parallax (ϖ)	194.95 ± 0.57	mas	van Leeuwen (2007)
Distance (d)	5.130±0.015	pc	van Leeuwen (2007)*
μ_α	536.024 ± 0.0420	mas yr^{-1}	Munn et al. (2022a)
μ_δ	385.173 ± 0.0560	mas yr^{-1}	Munn et al. (2022a)
U	-29.15 ± 0.31	km s^{-1}	this work
V	-10.43 ± 0.19	km s^{-1}	this work
W	-2.55 ± 0.18	km s^{-1}	this work

NOTE—[a]As observed from vantage point of JWST at the epoch of observation. * Distance d calculated as $1/\varpi$. Galactic Cartesian U, V, W velocity vectors in barycentric frame, with U towards Galactic center, V in direction of Galactic rotation, W towards North Galactic Pole, calculated using *astropy SkyCoord*.

3. DATA REDUCTION AND POST-PROCESSING

3.1. *Data Reduction*

Basic data reduction is done following Ygouf et al. (2024). For each telescope pointing, the images are flat-fielded, background subtracted, flagged for cosmic rays, co-added over multiple exposures, flux calibrated, and distortion corrected. At this level-2 stage of processing, images at the two telescope roll angles have not yet been combined, nor has the reference star been subtracted.

The data was processed with pipeline version 2025_2, calibration version 1.9.6, CRDS context for reference files jwst_1413.pmap, and distortion reference file jwst_nircam_distortion_0173.asdf. The dataset can be obtained at: https://doi.org/10.17909/61yw-dq54.

We have made some adjustments to the standard pipeline, to remove instrumental artifacts such as bad pixels and striping resulting from 1/f noise. While the standard pipeline flags pixels neighboring saturation-flagged pixels, we do not see any evidence of charge spillage and have removed those flags. **We use the overarching "DO_NOT_USE" pipeline flag to identify and NaN out other bad pixels (e.g. dead pixels, hot pixels, those without a linearity correction, etc.).**

As the final pre-processing step, each data cube of 650 individual frames was collapsed to a single median frame prior to PSF subtraction. Before this data stacking we remove pixels that are 5-σ outliers temporally (an outlier within one of the 650 sub-exposures) and then we remove outliers spatially (an outlier within a 5x5 box of surrounding pixels, after co-adding the sub-exposures).

3.2. *Post-Processing*

Point-spread function (PSF) subtraction was performed using the KLIP algorithm (Soummer et al. 2012) using the spaceKLIP (v2.2.1; Kammerer et al. 2022; Carter et al. 2023; Girard et al. 2022), which is a community supported python package designed for data reduction and analysis of JWST high-contrast imaging data. For the PSF subtraction algorithm, we used the pyKLIP (Wang et al. 2015) implementation of KLIP. Both epochs of our NIRCam data (PID 4534 and 8507) were reduced identically in our pre and post-processing pipelines. We performed the PSF subtraction using ADI (Marois et al. 2006; Liu 2004) with the median coad-

Andrea's notes

1 One of the first things we should look at after the abstract are tables and figures. Here we see a summary of the properties of Altair. Here is what each one means:
Spectral Type — type of star in our OBAFGKM categorization
Apparent V_mag — how bright the star appears to be relative to other stars in the V band (a particular range of wavelengths corresponding to visible light)
Apparent M_mag — how bright the star appears to be relative to other stars in the M band (another particular range of wavelengths in the infrared)
T_pole = temperature in the poles
T_eq = temperature in the equator
Mass = 1.86 times the solar mass
log(L/L_dot) = logarithm of the ratio of the star luminosity to the solar luminosity.
Age = 88 ± 10 Myr
[Fe/H] = ratio of iron to hydrogen relative to the Sun
log(g) = logarithm of gravity on the surface in cgs units
inclination(i) = how tilted it spins relative to our line of sight
theta_pole = how big the star looks in the sky pole to pole
theta_eq = how big the star looks in the sky along the equator
Next six values are the sky position of the star in the J2000 epoch in literature and the two JWST epochs for this work - you can see it shift. The units are Right Ascension and Declination which is the equivalent to latitude and longitude in the sky.
Parallax = small yearly wobble in its position as we see it as the Earth orbits the Sun. This helps us estimate its distance (d) in parsecs.
mu_alpha and mu_delta = proper motion of the star (aka how much it drifts per year across the sky)
U, V, W = its 3D velocity through the galaxy relative to the Sun. U = toward the galactic center, V = along the direction of galactic rotation, W = toward the galactic north pole

Table 2. NIRCam Coronagraphic Imaging Observing Parameters

Target	Filter	Readout	Groups/Int	Ints/Exp	Dithers	Total Time (s)
Epoch 1 - 2024 Oct 11						
Altair (Roll 1)	F200W/MASK430R	SUB320/RAPID	5	650	1	4182
Altair (Roll 1)	F444W/MASK430R	SUB320/RAPID	5	650	1	4182
Altair (Roll 2)	F200W/MASK430R	SUB320/RAPID	5	650	1	4182
Altair (Roll 2)	F444W/MASK430R	SUB320/RAPID	5	650	1	4182
Vega	F200W/MASK430R	SUB320/RAPID	5	140	5	901
Vega	F444W/MASK430R	SUB320/RAPID	5	140	5	901
Epoch 2 - 2025 May 26						
Altair (Roll 1)	F200W/MASK430R	SUB320/RAPID	5	650	1	4182
Altair (Roll 1)	F444W/MASK430R	SUB320/RAPID	5	650	1	4182
Altair (Roll 2)	F200W/MASK430R	SUB320/RAPID	5	650	1	4182
Altair (Roll 2)	F444W/MASK430R	SUB320/RAPID	5	650	1	4182
Vega	F200W/MASK430R	SUB320/RAPID	5	140	5	901
Vega	F444W/MASK430R	SUB320/RAPID	5	140	5	901

NOTE—Roll angle separation between Roll 1 and Roll 2 was 8.61° (Epoch 1) and 8.00° (Epoch 2). Both targets were acquired using F335M with the neutral density square. Vega served as the PSF reference star for RDI in both epochs, with 5 dithers for PSF diversity.

ded frames at two rolls, RDI (Lafrenière et al. 2009) using Vega as the reference star, and ADI+RDI which uses both. We chose to do the PSF subtraction with a single annulus and subsection (meaning full-frame subtraction). We performed the PSF subtraction for several KL modes as is standard in spaceKLIP including 1, 2, 5, and 10 modes. Figure 1 shows the final post-processed images of Altair with 5 KL modes at both F444W and F200W, for the two epochs.

4. RESULTS

4.1. *Sensitivity limits*

Contrast curves were computed using the spaceKLIP `AnalysisTools` module from the ADI+RDI PSF-subtracted images for 5 KL modes. Raw 5σ contrast curves were first calculated by computing the noise in azimuthal annuli at each separation and normalizing to the synthetic peak flux of the stellar PSF, estimated from a Vizier VOTable of Altair photometry[11] fit to a spectral type of A7V using `STPSF`. The raw contrast was then calibrated via fake companion injection-recovery tests to account for KLIP algorithmic throughput losses, with companions injected at separations of $0\farcs5$, $1''$, $2''$, $3''$, and $4''$ and position angles of 45° and 90° at a flux level of 20σ. We additionally corrected for small sample statistics at the closest separations (Mawet et al. 2014) and accounted for coronagraphic mask throughput losses. Calibrated contrast curves for both epochs in F444W and F200W are presented in Figure 2 (left).

Figure 2 (right) converts the F444W curves to planet mass limits using the Sonora Flame Skimmer (cloud-free) atmospheric and evolutionary models (Mang et al. 2026), which extend to sub-Saturn masses whereas many first-generation models (e.g., COND; Baraffe et al. 2003) reach their lower boundary at $\sim$0.5 $M_{\rm Jup}$. Specifically, we use the [M/H] = +0.5 dex metallicity, solar C/O ratio, $\log K_{zz} = 7$ dex models as a representative planet scenario. Note that mass limits are model-dependent and alternate atmospheric scenarios are discussed in Appendix A. We adopt Altair's age, distance, and F444W magnitude from Table 1 (Rieutord et al. 2024; van Leeuwen 2007). We do not convert F200W contrast to planet mass as it is primarily used for background star discrimination, as young planets emit negligibly at 2 μm relative to an A-type host. Interior to $\sim2''$ ($\sim$10 au), sensitivity is limited by coronagraphic mask throughput and residual PSF subtraction artifacts. In the background-limited regime beyond this radius, we achieve a maximum contrast of $\sim$19 mag, corresponding to $\sim$0.7 $M_{\rm Sat}$. We find no companion candidates with common proper motion with Altair, placing meaningful constraints on the giant planet population around Altair and making this one of the most sensitive direct imaging searches for low-mass companions around a nearby A-type star[12].

[11] Altair data were retrieved from Vizier on 2024-10-12, with a search radius of 1". One photometric point was excluded (Johnson:K, Morel & Magnenat 1978) due to anomalously large error bars.

[12] Beichman et al. (2025a) achieved similar contrast around Vega with JWST/NIRCam data, however the mass limits presented in this work are deeper due to the younger age of Altair.

Andrea's notes

1 It's important for every scientific paper to be replicable, so the parameters used to observe Altair are included here. This is not particularly relevant unless you are planning to request time on JWST to conduct similar analysis.

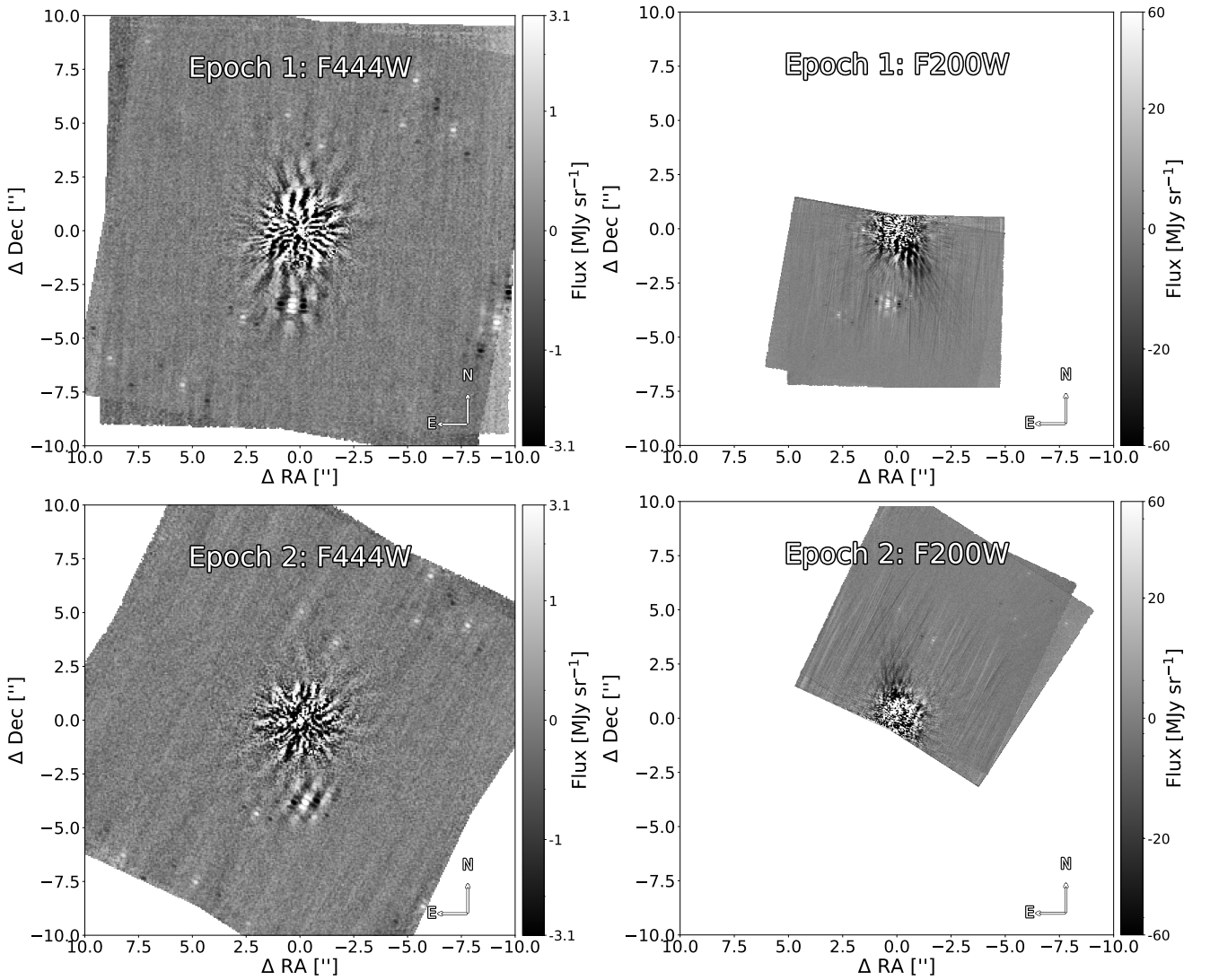


Figure 1. Post-processed images of Altair (ADI+RDI, 5 KL modes) with NIRCam. The top row shows Epoch 1 in F444W (left) and F200W (right); the bottom row shows Epoch 2 in the same filters. North is up, east is to the left. Point sources around Altair were shown to be background stars using a combination of F200W-F444W colors and lack of common proper motion with Altair in the second Epoch data. The negative sources on the images are due to the ADI subtraction.

4.2. *Point Sources Detected near Altair*

In the first-epoch F444W images (Figure 1, top left), we identified many point sources within 10″ of Altair by visual inspection. Sources to the South were quickly confirmed to be background stars based on their brightness in F200W (where planets are expected to be fainter) and F200W-F444W color. The selected mask (M430R) limits the field of view in the F200W image to the southern half of Altair, leaving any Northern point sources as companion candidates. These northern candidates lie at separations of $\sim 4''$ and have contrasts of $\sim$17.4 mag relative to Altair. If gravitationally bound, they would correspond to Saturn-mass companions or less at projected separations of 20-30 au. In our second-epoch observations, taken $\sim$7 months later, we were able to show that the candidate companions to the North are background stars, due to their lack of common proper motion with Altair, and their F200W-F444W colors (Figure 1, bottom).

4.3. *Bright Background Stars*

As Altair has no detected companions and no cold debris disk (Thureau et al. 2014; Spalding et al. 2022), it is a promising candidate reference star for future high-contrast coronagraphic imaging campaigns from the ground or space. We therefore present the position and brightness of 11 background stars in our image to support future observation planning. The source properties measured of the brighest background star due south of Altair data are summarized in Table 3, the other ten

Andrea's notes

1 Here we see the images of the two observations of Altair (the two epochs) in the two filters (F444W and F200W). These images have gone through some initial processing - so reduction like subtracting the background, removing cosmic rays or bad pixels.

ADI stands for angular differential imaging, and means that the two observations were done at different angles. The speckle pattern you see is due to the telescope optics, so will rotate with the telescope. Real objects in the image will stay where they are, so they will appear shifted by the same angle around the speckles. When you then try to subtract one image from the other to see the difference or remainder every object will appear as a bright dot and have a dark copy where it was in the other angle position. This is what it means by negative sources.

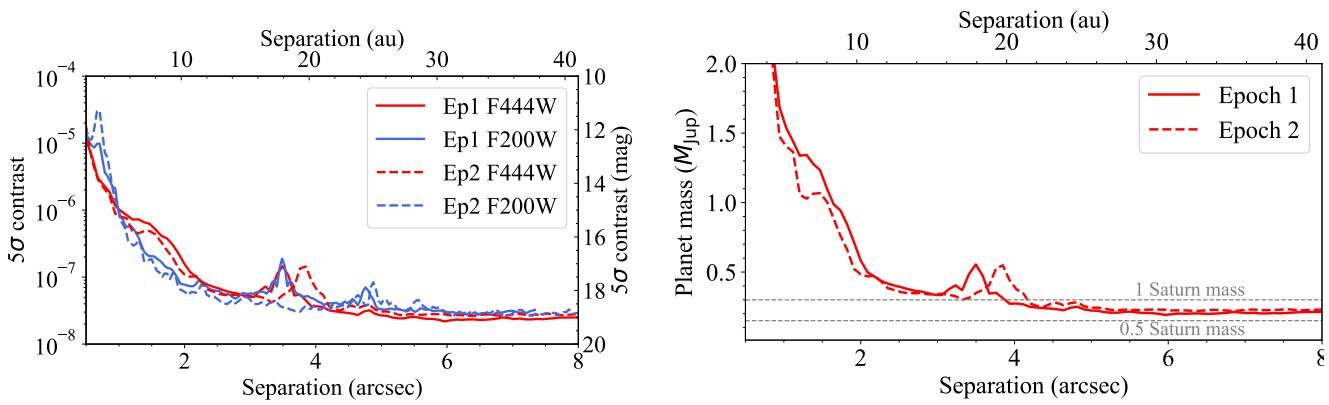


Figure 2. Sensitivity achieved in the JWST/NIRCam Altair data, showing deep limits on solar system orbital scales. *Left:* Contrast limits for the first epoch F444W JWST/NIRCam Altair data. The bump in contrast around 3.5" corresponds to the bright star south of Altair, visible in Figure 1. *Right:* Mass limits for the F444W NIRCam data computed using the [M/H] = +0.5 dex metallicity, solar C/O ratio, $\log K_{zz} = 7$ dex Sonora Flame Skimmer model (Mang et al. 2026), showing sensitivities down to 0.7 Saturn mass.

Table 3. Astrometry and photometry of the brightest background star (ID 1) south of Altair

ID	Date	Filter	ΔRA (″)	ΔDec (″)	PA (°)	Sep. (″)	Contrast
	2024 Oct 11	F200W	$+0.431 \pm 0.001$	-3.468 ± 0.001	172.9 ± 0.01	3.495 ± 0.001	$3.92 \pm 0.03 \times 10^{-7}$
1	2024 Oct 11	F444W	$+0.398 \pm 0.001$	-3.485 ± 0.001	173.5 ± 0.02	3.507 ± 0.001	$4.96 \pm 0.04 \times 10^{-7}$
	2025 May 26	F444W	-0.226 ± 0.001	-3.807 ± 0.001	183.4 ± 0.02	3.814 ± 0.001	$4.89 \pm 0.04 \times 10^{-7}$

Note. Astrometry and contrast measured from JWST/NIRCam coronagraphic imaging using `spaceKLIP`. Positive ΔRA is East, positive ΔDec is North. The source is confirmed as a background star based on its motion relative to Altair between epochs and its F200W-F444W color (Table 7). For the 10 less bright candidate sources identified in the field, see Figure 4 and Table 6 in the Appendix.

fainter objects identified in the field are described in Figure 4 and Table 6 in the Appendix. The astrometry and photometry for all point sources were measured using spaceKLIP's *extract_companions* function, which forward-models the JWST PSF and jointly fits position and flux via MCMC (Table 3 and 6).

We note that Altair is so nearby that its large angular diameter (∼ 3–3.7 mas; Monnier et al. 2007; Bourges et al. 2017; van Belle et al. 2001) may result in it being partially resolved by some telescopes. This angular diameter exceeds the < 2 mas ideal Roman Coronagraph Instrument reference star diameter (Hom et al. 2026). However, as the best reference stars for Roman are still being determined, we present this information for the Roman Coronagraph community and any other future high-contrast imaging campaigns where partial resolution of the stellar photosphere is not a limiting factor.

In the first epoch data, we identified a bright point source located almost due south of Altair at a separation of ∼3.5″. Due to the high proper motion of Altair (Table 1), our second epoch observations just ∼7 months later show the bright source moves relative to Altair by ∼700 mas, confirming it is a stationary background star via proper motion. This is further supported by its F200W-F444W color ($+0.27 \pm 0.07$; Table 7), which is consistent with a background star rather than the much redder colors expected of a young, self-luminous planet. Additionally, the point sources we identified in the field of view (Table 6) and show that they all have consistent apparent motion relative to Altair (Table 7), both in on sky motion (∼0.64–0.76″) and direction, further confirming their status as background stars rather than physically associated companions.

We additionally note that, looking ahead to future coronagraphic imaging campaigns which may use Altair as a reference star, the sky background in the vicinity of Altair will be relatively uncontaminated by background stars following the passage of this bright source (see Figure 1). Of the point sources identified in this work (Table 6), only objects 7 and 11 lie in the direction of Altair's proper motion. The other sources will continue to become more distant in projected separation. Object 7 will make its closest approach at ∼4.1″ separation around late 2029 (contrast ∼7.4×10^{-8} in

Andrea's notes

1 Simplifying the plots:

The left one tells us that as we separate from Altair (at the bottom you have the separation in arcseconds and at the top the equivalent separation in AU) we can detect fainter and fainter objects. 5sigma means we only count a signal if it is 5 times brighter than the typical noise at that point. So we can detect something a few hundred-millionths fainter than Altair at around 20AU out. On the right you have the mass of a detectable planet in Jupiter masses on the vertical axis, and what could be detected as you move farther out. So at around 20 - 30AU you can detect planets within 1-0.5 times Saturn's mass (or around 0.25 times Jupiter's mass!

The bump you see in both curves is due to a background star that makes it harder to spot faint objects.

2 We just learned there was a background star that caused the bump in contrast around Altair - these are the properties of that star.

F444W), and object 11 at ∼3.1″ around late 2040 (contrast ∼5.8×10^{-8} in F444W). Both are about an order of magnitude fainter than the bright background star characterized in Table 3, and neither will pass closer than ∼3.1″, suggesting a particularly favorable window for Altair as a reference star for high contrast imaging in the coming years.

5. DISCUSSION

Our deep NIRCam observations, sensitive to sub-Saturn-mass planets beyond 2″, reveal no confirmed companions and earlier observations revealed no debris disks emission. This contrasts with other A stars like Vega and β Pic which host either a disk (Smith & Terrile 1984; Aumann et al. 1984), planets (Lagrange et al. 2010) or both. Based on planet occurrence rates and mass functions presented in Meyer et al. (2025), even with a null detection in our deep dataset, $< 1 - 2M_{\rm Jup}$ planets may still reside within 10 au where occurrence rates are expected to be higher.

Altair is somewhat unusual but not exceptional in its lack of a disk and its high rotational velocity (v sin i =221 km s^{-1} (Erspamer & North 2003). The simplest explanation for these properties is that Altair is one of those A stars born with a high rotational velocity and which has not been appreciably slowed given its relatively youth and the inefficiency of magnetic braking in hot stars with thin or non-existent convective layers (Zorec & Royer 2012).

A more exotic explanation of the high rotational velocity is that Altair ingested a massive planet or low mass star with a consequent increase in its rotational angular momentum. The presence of such a companion on an eccentric orbit and eventually fatal trajectory could also clean out a residual debris disk and account for Altair's lack of a significant level of IR emission (Beichman et al. 2025b). Ingestion of a companion, particularly a planetary mass object, would be expected to change the star's elemental abundances (Behmard et al. 2022). Measuring elemental abundances for Altair is challenging given its rapid rotation ($v \sin i \sim 240$ km s^{-1}). The recent spectroscopic analysis by Takeda (2026) measuring the abundances of 17 elements finds that Altair is slightly subsolar ([Fe/H] ~ -0.2), differential abundances with respect to the Sun are consistent with $-0.5 <$ [X/Fe] $< +0.3$, lacking any signs of a dependence on atomic number, and no similarity to other peculiar classes (e.g. Am, lamba Boo stars). Hence, Altair appears to be a "normal" A-type star, not showing any abundance effects that might indicate evidence of past planetary engulfment.

Finally, while debris disks among young A stars are common, there are numerous examples of young A stars without detectable disks (Su et al. 2006). The lack of any planet could be attributed to rapid dissipation of the protostellar disk (Ribas et al. 2015).

5.1. *A Search for Altair Siblings*

As there is some disagreement in the literature on the age of Altair, and given the surprise of the extremely young age of 88 ± 10 Myr recently derived by Rieutord et al. (2024), we searched for any evidence of low-mass stellar companions or a moving group to which Altair might belong.

5.1.1. *Search for a Kinematic Group*

We calculate an updated space velocity for Altair based on combining the available astrometry and radial velocity estimates with the smallest uncertainties, including parallax from van Leeuwen (2007), proper motion and ICRS position from Munn et al. (2022a), and radial velocity from Gontcharov (2006). Based on these values, we estimate the Galactic Cartesian velocity of Altair (in the solar system barycentric frame) using *Astropy SkyCoord* to be $(U, V, W) = (-29.15, -10.43, -2.54) \pm (0.27, 0.29, 0.06)$ km s^{-1}.

The BANYAN Σ (Bayesian Analysis for Nearby Young AssociatioNs Σ) tool from Gagné et al. (2018) was used to search for potential host stellar association for Altair. The SIMBAD default values for astrometry from van Leeuwen (2007) and the literature mean radial velocity from Gontcharov (2006) were taken as inputs. BANYAN Σ estimated the probability of the star being "field" was 99.7% and attributed a 0.3% probability of membership to the 40 Myr-old Argus group, and a <0.1% membership probably to the other $d < 150$ pc stellar groups surveyed by Gagné et al. (2018).

Gagné et al. (2026) updated the BANYAN Σ model to included recent Gaia results related to the discovery of new kinematic groups and membership analyses, creating the Montreal Open Clusters and Associations database (MOCAdb). A query of MOCAdb using the astrometry from van Leeuwen (2007) and radial velocity from Kharchenko et al. (2007), yielded a 98.9% probability of Altair being a field star, with negligible probabilities that it could belong to Carina-Near ("CARN"; 0.75%) and [MGC2022] Crius 209 (0.33%). We discuss these two groups further to see whether there is any plausibility of an genetic tie with Altair.

The Carina-Near group is a ∼ 200 Myr-old small group discovered by Zuckerman et al. (2006) situated ∼30 pc from the Sun. Mamajek (2016) notes that group is very small, with no known members hotter than F1, an estimated total mass of only ∼ 8 M$_\odot$, and an intrinsic

Andrea's notes

1. Altair seems to be a bit peculiar, so it is helpful to see if any literature on similar stars can be found. Perhaps then we can compare properties to see if it belongs to a unique family of stars.
2. In general, I would recommend reading the abstract, figures, discussion and conclusions first.
3. It seems Altair is a bit of an outlier!
4. We saw this earlier, but they calculated exactly how fast Altair is moving through the sky.
5. This tells us that ther are no planets within 1-2 Jupiter masses, but there could be some less than this mass within 10AU which has not been probed in this analysis. If you look at the sensitivity curves in Figure 2 you can see that in the area less than 10 arcseconds, you can only really detect bright objects due to the leftover starlight, which would be objects with >1 Jupiter mass. However it says that literature suggests that smaller planets at these inner distances occur more frequently, and this study is not sensitive enough to determine if they exist.
6. A "field" star is one that doesn't belong to any cluster or group, and moves through the galaxy on its own. Models suggest a 99.7% probability that it is a field star.

   Stars are usually born together in groups or clusters, and travel together through the galaxy. So you can usually tell the members of a group from their similar movements.
7. The proportions of different elements the star contains - we would determine this using its spectrum.
8. Another check yielded a 98.9% probability of Altair being a lone star!
9. A ratio of 0 means that it is like the Sun in its composition of elements
10. This tells us that the elements of Altair do not show any evidence of it engulfing something like a planet - which was a hypothesis presented for why it had such a high rotation speed and no debris disk for such a young star.

1D velocity dispersion of $\sim$1.3 km s^{-1} (Mamajek 2016). The velocity of Car-Near ($U, V, W = -25.3, -18.1, -2.3$ km s^{-1}) differs from Altair by $\Delta S = 8.6$ km s^{-1}, at roughly $\sim 5.8\sigma$ significance if one considers the 1D velocity dispersion of Car-Near, uncertainties in mean UVW vector for Car-Near, and uncertainties in Altair's UVW. Altair is 32 pc from the Car-Near centroid (Gagné et al. 2018), and running the orbits of Altair and Carina-Near back in time using epicyclic approximation following Faramaz et al. (2021), we find that Altair was no closer than $\sim$15 pc from the Carina-Near centroid (3.4 Myr ago), and passed with relative velocity $\delta S \simeq 8.3$ km s^{-1}. Given the lack of more massive systems in the Carina-Near nucleus, an ejection would have required a more massive star ($\sim$ 4.2 M$_\odot$, if it ended its life 3 Myr ago within a $\sim$200 Myr-old cluster). However Altair shows no signs of anomalous elemental abundances that might suggest ejection from a close-in binary with a more massive partner.

Moranta et al. (2022) first identified the candidate group [MGC2022] Crius 209 in their Gaia EDR3 6D clustering analysis using HDBSCAN, and presented a membership list. However the group failed two of their four criteria for declaring a new association for which they would quote an age, instead Crius 209 had notably large scatter in Z and W. Moranta et al. (2022) notes "*that these rejected Crius groups may contain additional coeval structures, but they will likely also include a larger number of contaminated or unphysical clusters.*" The Crius 209 sample from Moranta et al. (2022) does not include any stars brighter than V or G of 7th magnitude, and their distances are $d \simeq 66 \pm 21$ pc (rms). Taking the Crius 209 centroid position and velocity from Moranta et al. (2022) at face value, the velocity of Crius 209 differs from that of Altair by $\Delta S = 7.1$ km s^{-1} ($\chi^2/\nu = 8.6/3$, P=0.036). Running the orbits of Altair and Crius 209 back in time using epicyclic approximation following Faramaz et al. (2021), we find that Altair came no closer than about 32 pc from the Crius 209 centroid, and had relative velocity $\Delta S = 6.8$ km s^{-1}. Again, there seems to be no smoking gun explanation for why a late A star like Altair would be ejected from such a small group, itself containing only one other A-type star, and the physicality of the group itself is in question.

Hence, we conclude that it seems exceedingly unlikely that Altair could have any genetic connection to either Carina-Near or the putative Crius 209 group.

5.1.2. *Washington Double Star Catalog*

Stellar companions have proved valuable for constraining the ages of some of the nearest A-type stars (e.g., Sirius, Fomalhaut; Bond et al. 2017; Mamajek 2012). Given that Altair lies at only 5 pc, it would have been remarkable indeed if a resolved stellar companion had been missed from recent previous surveys of nearby stars and wide companions (e.g. Kirkpatrick et al. 2024; González-Payo 2025).

The Washington Double Star Catalog (Mason et al. 2001)[13] has six double star entries for Altair (WDS 19508+0852). In Table 4 we summarize the WDS entries for Altair (component A). Components B and C were already cross-IDed in SIMBAD. Using the WDS position angle and separations, we cross-IDed components D through G which were missing in SIMBAD. In Table 5 we list the Gaia IDs, parallaxes, and proper motions for the WDS companions, and compare their values to those of Altair. The companions WDS 19508+0852B through G all have parallaxes and proper motions consistent with being distant ($>$ 0.4 kpc) background stars, and we conclude that none of the WDS entries associated with Altair are physical companions.

Table 4. WDS Double Stars Associated with Altair (WDS J19508+0852)

Disc.	Comps.	θ	ρ	mag
ID	...	$^\circ$	$''$	...
STFB 10	AB	286	195.80	9.82
STFB 10	AC	110	186.40	10.30
DAL 27	AD	105	26.80	11.90
SMR 5	AE	351	151.70	11.00
SMR 5	AF	48	292.40	10.30
GMC 5	AG	121	185.10	13.00

NOTE—Columns: (1) WDS discoverer ID, (2) component letters for WDS 19508+0852 (A = Altair), (3) position angle θ ($^\circ$), (4) separation ρ ($''$), (5) apparent magnitude in WDS (system not specified).

Thus far, our analysis is consistent with the notion that Altair is a single star that does not belong to any known stellar association, a perhaps surprising result given its young asteroseismic age.

6. CONCLUSIONS

We observed Altair with the JWST/NIRCam coronagraph in F444W and F200W, achieving contrast levels of 1.5×10^{-7} at 2" (10 au) and a few $\times 10^{-8}$ beyond 4" (20 au). Using its recently revised age of 88$\pm$10 Myr (Rieutord et al. 2024) and the Sonora Flame Skimmer model (Mang et al. 2026), this corresponds to sub-Saturn mass

[13] https://astro.gsu.edu/wds/

Andrea's notes

1 It is always possible that we missed a second star with Altair. However it is really close (on cosmic scales 5pc is nothing), so it is very unlikely that we somehow missed it.

2 I didn't know anything about this but apparently the WDS shows candidate companion stars. The team measured the motion of these stars and distances from the parallax and compared them to Altair. They had the motion they would expect from background stars and were shown to be at more than 0.4kpc away (bear in mind Altair is 5pc away and these are more than 100 times further! - 1kpc = 1000 pc)

3 The details are a bit technical for someone that is not in this field, but this paragraph was an attempt to see whether Altair belonging to Car-Near was plausible. The explanation would be that it was ejected from a binary system, but its element composition does not suggest this.

4 There was another group Altair could have been part of due to proximity, but again they conclude that there is no reason to explain why Altair would be in it and then ejected. Also the group 'definition' seems to be still in question.

5 Asteroseismology is the method used to show that Altair's age is 88Myr in the literature. It involves studying seismic activity (kinda like earthquakes!) through the star to study its properties such as age and mass.

6 Notice that the conclusion summarizes things that were said in the abstract - we didn't learn anything new here!

7 Stellar companions (aka binary stars/double stars or more) help us narrow down the properties of stars, because we have more information about them than if they were just alone.

Table 5. Cross-IDs for Altair's WDS entries

WDS ID	Alias	ϖ	μ_α	μ_δ	Ref.
...	...	(mas)	mas yr^{-1}	mas yr^{-1}	...
WDS J19508+0852A	Altair	194.95 ± 0.57	536.024 ± 0.042	385.173 ± 0.056	1,2
WDS J19508+0852B	Gaia DR3 4298633621839700736	2.0429 ± 0.0132	7.230 ± 0.012	-1.974 ± 0.010	3
WDS J19508+0852C	Gaia DR3 4298620668218220160	1.0635 ± 0.0175	-12.371 ± 0.014	-4.639 ± 0.013	3
WDS J19508+0852D	Gaia DR3 4298621664638734592	0.4832 ± 0.0166	-0.334 ± 0.013	-1.201 ± 0.012	3
WDS J19508+0852E	Gaia DR3 4298623520076617728	1.3878 ± 0.0153	-6.628 ± 0.014	-22.367 ± 0.012	3
WDS J19508+0852F	Gaia DR3 4298623416997887232	1.5098 ± 0.0225	-1.353 ± 0.018	-6.228 ± 0.017	3
WDS J19508+0852G	Gaia DR3 4298620633858473344	0.6359 ± 0.0209	1.891 ± 0.017	1.631 ± 0.015	3

NOTE—References: (1) van Leeuwen (2007), (2) Munn et al. (2022b), (3) Gaia Collaboration et al. (2023).

limits beyond 10 au. We identified several companion candidates in the first epoch data. Second epoch observations showed these candidates were not co-moving with Altair, and their colors consistent with background stars. Altair remains elusive, with no detected debris disk, companions, and it is not known to be part of any stellar association. Its revised young age and proximity makes it a compelling target for future high-contrast imaging observations. We also characterize the background star field around Altair and find that it will remain a few arcsec from any background sources over the coming decade, making it a reliable reference star for future coronagraphic imaging campaigns, particularly for the Roman Coronagraph Instrument.

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AUTHOR CONTRIBUTIONS

TM led the project, obtained the second epoch observations, performed the data analysis, and wrote and submitted the manuscript. CB came up with the initial research concept, led the NIRCam GTO team, wrote the discussion of Altair's lack of a detected disk, and edited the manuscript. JLS contributed to the post-processing analysis and edited the manuscript. GB performed the data pre-processing and edited the manuscript. AZG supported the data processing, provided expertise on NIRCam data analysis, and edited the manuscript. EM led the search for Altair co-moving siblings and the stellar age analysis, and edited the manuscript. MY supported the data processing and provided expertise on NIRCam data analysis. JG contributed to determining the optimal pointing the mask for the second epoch based on the first epoch. JL contributed to determining the optimal pointing the mask for the second epoch based on the first epoch. AS provided expertise on converting the achieved contrast to planet masses and edited the manuscript. MP, MM, LP, WB, AG, and KH contributed as a member of the NIRCam GTO team. All authors assisted in the preparation of the second epoch JWST proposal.

Facilities: JWST (NIRCam)

Software: astropy (Astropy Collaboration et al. 2013, 2018, 2022), spaceKLIP (Kammerer et al. 2022; Carter et al. 2023), BANYAN (Gagné et al. 2018), MOCA (Gagné et al. 2026).

APPENDIX

A. PLANET MASS LIMITS

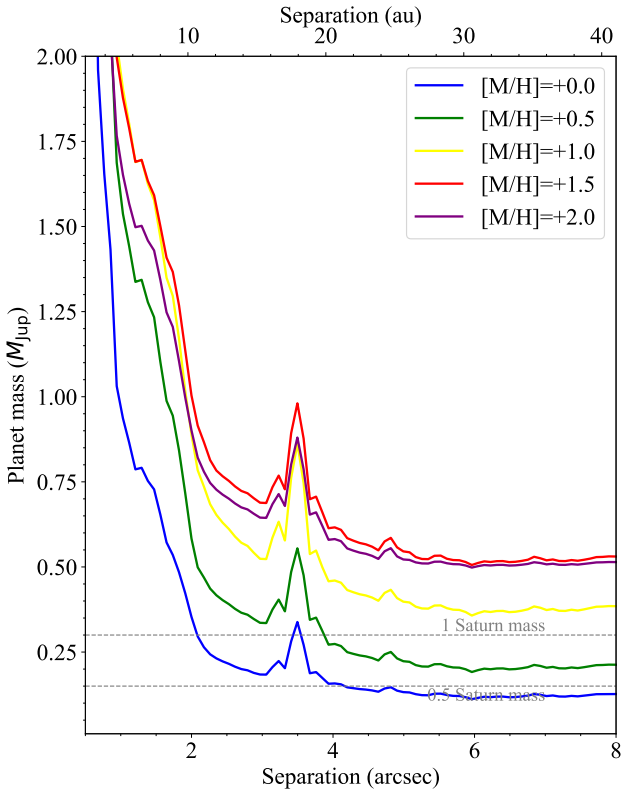


Figure 3. A comparison of the planet mass sensitivity achieved by the NIRCam F444W Altair observations for varying atmospheric metallicities scenarios with the Sonora Flame Skimmer models (assuming strong vertical mixing and solar C/O ratio).

The conversion of the F444W contrast curve to a planetary mass sensitivity curve relies on the use of atmospheric and evolutionary models and requires assumptions to be made about the planet's atmospheric parameters, such as metallicity, vertical mixing strength, C/O ratio, and the presence/absence of clouds. A growing number of JWST observations and modeling efforts are now providing evidence that the 4–5 μm (F444W) flux of cold ($\lesssim$400 K), low-mass giant planets (such as the ones we are sensitive to in this work) can be dramatically lower due to a combination of strong vertical mixing and enhanced metallicity, which increase the strength of CO and CO_2 absorption, and the presence of water ice clouds, which act as an additional opacity source by redistributing flux to longer wavelengths (e.g., Morley et al. 2014; Lacy & Burrows 2023; Crotts et al. 2025; Bardalez Gagliuffi et al. 2025; Bowens-Rubin et al. 2025; Sanghi et al. 2026a,b).

To account for this atmospheric model dependence, we additionally compute mass sensitivity curves for varying atmospheric metallicity (0 dex to +2.0 dex) and with strong vertical mixing (conservative case) using the clear Sonora Flame Skimmer models (Mang et al. 2026). Figure 3 shows that the mass sensitivity degrades as the atmospheric metallicity increases, as expected. However, in all cases, we are still sensitive to $< 1\ M_{\rm Jup}$ planets at separations $\gtrsim 2''$. While the Sonora Flame Skimmer models do not include the effects of water clouds at cold temperatures, the highest metallicity case (+2.0 dex) can be used as a representative proxy since the impact on the F444W flux is similar. Note that Jupiter's atmosphere is enriched in metals at 3$\times$ (+0.5 dex) the solar value (Wong et al. 2004) and Saturn's

atmosphere is enriched at the 10× (+1.0 dex) the solar value (Fletcher et al. 2009), so it is unlikely that any giant planet around Altair is enriched at the +2.0 dex level.

B. BACKGROUND STAR ASTROMETRY AND PHOTOMETRY

In addition to the bright background star discussed in Section 4.3 (Table 3), ten fainter point sources were identified in the field of view around Altair. This appendix presents their measured astrometry and photometry from spaceKLIP's *extract_companions* function (Table 6), their F200W-F444W colors where simultaneous multi-filter coverage is available (Table 7), and their locations relative to Altair (Figure 4).

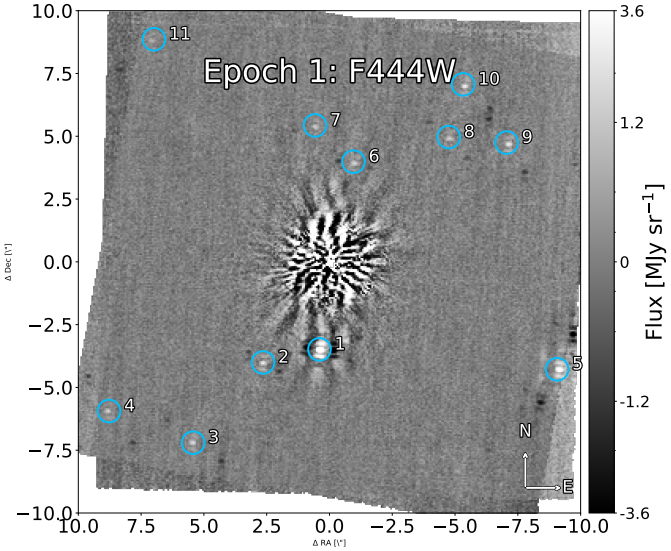


Figure 4. KLIP-subtracted NIRCam coronagraphic images of the Altair field across both epochs and filters, with the eleven identified point sources circled and numbered. Object 1 is the bright background star discussed in the main text (Table 3); objects 2–11 are presented in Table 6. Not every object falls within the field of view of every epoch/filter combination.

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Table 6. Astrometry and photometry of additional candidate point sources in the Altair field. See Table 3 for the brightest background star (object 1).

ID	Date	Filter	ΔRA (″)	ΔDec (″)	PA (°)	Sep. (″)	Contrast
2	2024 Oct 11	F200W	$+2.6727 \pm 0.0018$	-3.9933 ± 0.0011	146.21 ± 0.02	4.8052 ± 0.0013	$9.69 \pm 0.20 \times 10^{-8}$
	2024 Oct 11	F444W	$+2.6445 \pm 0.0036$	-4.0039 ± 0.0024	146.56 ± 0.04	4.7984 ± 0.0028	$1.37 \pm 0.03 \times 10^{-7}$
	2025 May 26	F444W	$+2.0394 \pm 0.0038$	-4.3350 ± 0.0033	154.81 ± 0.04	4.7907 ± 0.0034	$1.06 \pm 0.03 \times 10^{-7}$
3	2024 Oct 11	F444W	$+5.4262 \pm 0.0067$	-7.2008 ± 0.0047	143.00 ± 0.04	9.0164 ± 0.0055	$1.20 \pm 0.04 \times 10^{-7}$
	2025 May 26	F444W	$+4.8433 \pm 0.0057$	-7.5137 ± 0.0052	147.19 ± 0.04	8.9395 ± 0.0053	$8.07 \pm 0.35 \times 10^{-8}$
4	2024 Oct 11	F444W	$+8.7754 \pm 0.0244$	-5.9408 ± 0.0181	124.10 ± 0.11	10.5972 ± 0.0226	$6.69 \pm 1.08 \times 10^{-8}$
	2025 May 26	F444W	$+8.2051 \pm 0.0210$	-6.2746 ± 0.0192	127.41 ± 0.11	10.3293 ± 0.0204	$8.41 \pm 1.21 \times 10^{-8}$
5	2024 Oct 11	F444W	-9.0636 ± 0.0066	-4.2770 ± 0.0052	244.74 ± 0.03	10.0221 ± 0.0064	$1.89 \pm 0.11 \times 10^{-7}$
6	2024 Oct 11	F444W	-0.9580 ± 0.0054	$+3.9877 \pm 0.0036$	346.49 ± 0.07	4.1011 ± 0.0037	$1.13 \pm 0.04 \times 10^{-7}$
	2025 May 26	F444W	-1.6311 ± 0.0076	$+3.6306 \pm 0.0060$	335.81 ± 0.11	3.9802 ± 0.0063	$1.00 \pm 0.05 \times 10^{-7}$
	2025 May 26	F200W	-1.6513 ± 0.0021	$+3.6172 \pm 0.0017$	335.46 ± 0.03	3.9763 ± 0.0018	$6.43 \pm 0.19 \times 10^{-8}$
7	2024 Oct 11	F444W	$+0.5721 \pm 0.0059$	$+5.4323 \pm 0.0036$	6.01 ± 0.06	5.4623 ± 0.0037	$1.03 \pm 0.03 \times 10^{-7}$
	2025 May 26	F444W	-0.0504 ± 0.0087	$+5.0926 \pm 0.0069$	359.43 ± 0.10	5.0928 ± 0.0069	$4.67 \pm 0.29 \times 10^{-8}$
	2025 May 26	F200W	-0.0951 ± 0.0058	$+5.0586 \pm 0.0050$	358.92 ± 0.07	5.0595 ± 0.0050	$1.90 \pm 0.18 \times 10^{-8}$
8	2024 Oct 11	F444W	-4.7495 ± 0.0058	$+4.9669 \pm 0.0044$	316.28 ± 0.04	6.8722 ± 0.0051	$8.73 \pm 0.28 \times 10^{-8}$
	2025 May 26	F444W	-5.3676 ± 0.0058	$+4.6723 \pm 0.0047$	311.04 ± 0.04	7.1162 ± 0.0053	$8.09 \pm 0.27 \times 10^{-8}$
	2025 May 26	F200W	-5.4187 ± 0.0035	$+4.6191 \pm 0.0030$	310.45 ± 0.03	7.1203 ± 0.0033	$3.08 \pm 0.14 \times 10^{-8}$
9	2024 Oct 11	F444W	-7.0564 ± 0.0052	$+4.7487 \pm 0.0035$	303.94 ± 0.03	8.5055 ± 0.0047	$8.95 \pm 0.35 \times 10^{-8}$
	2025 May 26	F444W	-7.7285 ± 0.0040	$+4.4462 \pm 0.0032$	299.91 ± 0.02	8.9162 ± 0.0038	$1.33 \pm 0.03 \times 10^{-7}$
	2025 May 26	F200W	-7.7626 ± 0.0014	$+4.4104 \pm 0.0014$	299.60 ± 0.01	8.9280 ± 0.0014	$1.19 \pm 0.03 \times 10^{-7}$
10	2024 Oct 11	F444W	-5.3220 ± 0.0049	$+7.0661 \pm 0.0034$	323.01 ± 0.03	8.8461 ± 0.0040	$1.08 \pm 0.04 \times 10^{-7}$
	2025 May 26	F444W	-6.0054 ± 0.0051	$+6.7539 \pm 0.0042$	318.36 ± 0.03	9.0377 ± 0.0046	$1.00 \pm 0.04 \times 10^{-7}$
	2025 May 26	F200W	-6.0328 ± 0.0058	$+6.7122 \pm 0.0044$	318.05 ± 0.03	9.0249 ± 0.0051	$7.28 \pm 0.42 \times 10^{-8}$
11	2024 Oct 11	F444W	$+6.9878 \pm 0.0346$	$+8.8614 \pm 0.0381$	38.26 ± 0.18	11.2851 ± 0.0368	$4.34 \pm 1.08 \times 10^{-8}$
	2025 May 26	F444W	$+6.4505 \pm 0.0471$	$+8.5213 \pm 0.0366$	37.13 ± 0.23	10.6875 ± 0.0407	$1.10 \pm 0.20 \times 10^{-7}$

Note. Astrometry and contrast measured from JWST/NIRCam coronagraphic imaging using `spaceKLIP`. Positive ΔRA is East, positive ΔDec is North. Objects 2–4 and 6–10 show consistent apparent motion relative to Altair between epochs (∼0.7″ over ∼7.5 months, in a common direction), confirming their identification as background stars rather than physically associated companions. Objects 3 and 4 fall outside the field of view of the F200W Epoch 1 observation. Object 5 is detected in only one epoch/filter combination and its background/companion status remains unconstrained. Object 11's motion is broadly consistent with the background-star group, though its low SNR (∼2–5) makes this less certain.

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Table 7. Background object F200W−F444W colors and apparent motion relative to Altair between epochs

ID	F200W−F444W	Motion (″, PA)
1	$+0.27 \pm 0.07$	0.70, 242.7°
2	$+0.39 \pm 0.08$	0.69, 241.3°
3	...	0.66, 241.8°
4	...	0.66, 239.7°
6	$+0.50 \pm 0.10$	0.76, 242.1°
7	$+0.99 \pm 0.14$	0.71, 241.4°
8	$+1.06 \pm 0.10$	0.68, 244.5°
9	$+0.14 \pm 0.08$	0.74, 245.8°
10	$+0.37 \pm 0.11$	0.75, 245.5°
11	...	0.64, 237.7°

Note. Colors are only computed when there is simultaneous F200W and F444W measurements. Objects 3, 4, and 11 do not have simultaneous F200W/F444W coverage due to the limited FOV of F200W. Object 5 (only detected in Epoch 1, F444W) is removed from this table, as no motion or color measurement could be computed. "Motion" is the magnitude and position angle (E of N) of each object's apparent motion relative to Altair between the two epochs (∼7.5 months apart), derived from Table 6. The spread in color (+0.14 to +1.06 mag) is consistent with a population of background stars of varying spectral type. The consistent motion across all sources (0.64-0.76″, 238–246°) confirms their status as stationary background objects rather than physically associated companions.

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